

Notes: Santacreu (JPE, 2025)

Dynamic Gains from Trade Agreements with Intellectual Property Provisions

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1 Big picture

- **Question.** What are the dynamic gains and losses from a bilateral trade agreement that jointly negotiates **tariffs** and **intellectual property (IP) protection**, especially between a developed innovating country and a developing adopting country?
- **Setting.** The paper studies a multicountry trade-and-growth model calibrated to the **United States, China, and the rest of the world**. Productivity growth is driven by **innovation** and **international technology licensing** measured with royalty payments.
- **Core challenge.** Trade agreements with IP provisions do not only change static trade flows. They also alter incentives to innovate, adopt foreign technologies, and pay royalties. That means the relevant welfare object is **dynamic welfare along the transition**, not just a before-versus-after balanced-growth comparison.
- **Main theoretical result.** Weak IP enforcement lowers royalty payments, depresses innovators' returns, and reduces long-run growth. A deep trade agreement can correct this inefficiency by combining **lower tariffs** with **stronger IP enforcement**.
- **Main quantitative result.** In the baseline Nash-bargaining agreement, the United States removes tariffs on Chinese imports and China improves IP enforcement for both domestic and foreign technologies. Welfare rises in all countries overall, but the transition is uneven: the **United States gains immediately**, while **China suffers short-run losses** and gains only later through faster growth.
- **Main empirical motivation.** Royalty payments from developing to developed countries rise sharply after regional trade agreements that include **IP provisions**, but much less after agreements without such provisions.
- **Why the paper matters.** It connects the political economy of trade agreements to quantitative models of trade, innovation, and diffusion. The key message is that deep trade agreements are about **growth, licensing, and transition dynamics**, not only about static market access.

2 Data and measurement (key objects)

2.1 Observed bilateral royalty-payment panel

- **Core licensing data.** Bilateral royalty payments come from the OECD **Balanced International Trade in Services** data under EBOPS 2012.
- **Main time coverage for motivation.** The event-study evidence uses bilateral royalty-payment data for **50 countries from 1995 to 2012**.
- **Economic role of royalties.** The paper treats royalty payments as the main observable measure of **international technology licensing** and therefore of one key channel of technology transfer.
- **Country direction of interest.** The focus is on royalty flows from **developing to developed countries**, because trade agreements with IP provisions are argued to matter most in that direction.

2.2 Observed trade-agreement panel

- **Regional trade agreements.** The paper uses a dataset of RTAs with and without IP provisions from Martínez-Zarzoso and Chelala (2021).
- **Agreement split.** The motivating figure compares pairs that sign **RTAs with technology/IP provisions** against pairs that sign **RTAs without such provisions**.
- **Key measured timing.** Royalty payments are normalized to one in the year the agreement enters into force and then tracked several years before and after implementation.

2.3 Observed trade, geography, and macro panels for calibration

- **Trade-flow data.** Bilateral trade, including international and intranational trade, are used to estimate trade costs and relative productivity through a gravity equation. The calibration uses **69 countries over 1986–2006**.
- **Geography and size.** Population and GDP data are used from standard cross-country sources such as CEPII.
- **IP-rights index.** The paper uses the **Ginarte-Park patent-rights index** to back out the quality of IP enforcement from royalty-payment data.
- **R&D intensity targets.** Country-level R&D intensities around 2000 help pin down innovation parameters in the model.

2.4 Key measured objects

- **Royalty payments** as technology licensing flows.
- **Trade shares** and **home trade shares** as the static trade outcomes affected by tariffs and trade costs.
- **Relative productivity** across countries, inferred from trade gravity.
- **IP enforcement quality**, inferred from royalty gravity and patent-rights data.
- **R&D intensity**, used to pin down innovation efficiency and growth.

2.5 Unobservables and timing

- **Technology stocks.** $Z_{n,t}$ is the stock of innovations in country n , $A_{in,t}$ is the stock of technologies from n adopted by i , and $T_{n,t}$ is the total number of technologies used in production in n .
- **Innovation and adoption efficiency.** Country-specific parameters ℓ_n and $\bar{\varepsilon}_{in}$ govern how easily countries innovate and adopt technologies.
- **IP enforcement quality.** The parameter h_i captures how much of the bargained royalty innovator actually receives when country i is the adopter.
- **Timing.** In the baseline counterfactual, the economy starts on an initial BGP. In period 1, the trade agreement is signed as an **unexpected, permanent, perfectly enforced shock**, and the model is solved with perfect foresight along the transition.

2.6 Empirical and quantitative applications

- Event-study evidence on royalty payments around RTAs with IP provisions.
- Structural gravity estimation for trade costs, productivity, adoption probabilities, and IP enforcement.
- Baseline Nash-bargaining design of a US-China trade agreement.
- Counterfactual analysis of tariff-only reforms, IP-only reforms, unilateral IP reform, uncooperative equilibrium, shortsighted governments, deviation and retaliation, and anticipated or gradual implementation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Preferences and household budget constraint

In each country n , a representative consumer chooses consumption $C_{n,t}$ to maximize

$$\sum_{t=0}^{\infty} \beta^t \log(C_{n,t}),$$

subject to

$$P_{n,t}C_{n,t} + P_{n,t}B_{n,t} + \frac{h}{2}(B_{n,t} - \bar{B}_n)^2 = W_{n,t}L_{n,t} + \Pi_{n,t}^{\text{all}} + R_t P_{n,t}B_{n,t-1} + IBT_{n,t} + Tr_{n,t}.$$

- **Interpretation.** Consumers receive wages, firm profits, tariff rebates, and bond income, and they finance current consumption plus bond holdings.
- **Why the quadratic bond term matters.** The term $\frac{h}{2}(B_{n,t} - \bar{B}_n)^2$ is a standard stationarity device that pins down a unique steady state for international bond holdings.

3.2 Final-good production and demand for intermediates

The final good in country n is produced competitively using a CES aggregator of traded intermediate goods:

$$Y_{n,t} = \left[\sum_{i=1}^M \int_0^{T_{i,t}} x_{ni,t}(j)^{(\eta-1)/\eta} dj \right]^{\eta/(\eta-1)},$$

where $\eta > 1$ is the elasticity of substitution across intermediates.

Demand for intermediate good j from country i in country n is

$$x_{ni,t}(j) = \left(\frac{p_{ni,t}(j)}{P_{n,t}} \right)^{-\eta} Y_{n,t}.$$

- **Interpretation.** This is an Armington-style trade block with differentiation by both variety and country of origin.

- **Why $T_{i,t}$ matters.** The mass of varieties available from country i is endogenous because it depends on past innovation and adoption.

3.3 Intermediate-good production and trade barriers

Each intermediate producer in country n uses labor linearly:

$$y_{n,t}(j) = \Omega_n \ell_{n,t}(j),$$

where Ω_n is country- n productivity.

Profits for an intermediate producer in country n are

$$\pi_{n,t}(j) = \sum_{i=1}^M p_{in,t}(j) x_{in,t}(j) - W_{n,t} \ell_{n,t}(j).$$

Trade is subject to both iceberg trade costs and tariffs. The import share of country n from country i is

$$\pi_{ni,t} = \frac{T_{i,t} \Omega_i^{\eta-1} (W_{i,t} d_{ni} (1 + t_{ni,t}))^{1-\eta}}{\sum_{m=1}^M T_{m,t} \Omega_m^{\eta-1} (W_{m,t} d_{nm} (1 + t_{nm,t}))^{1-\eta}}.$$

- **Interpretation.** Trade shares depend on variety counts, productivity, wages, iceberg trade costs, and tariffs.
- **Economic channel.** Lower tariffs on Chinese goods reduce the US home trade share and create standard static gains from trade.

3.4 Innovation

A monopolist in country n invests final output $H_{n,t}^r$ to create new technologies according to

$$Z_{n,t+1} = \ell_n T_{n,t} \left(\frac{H_{n,t}^r}{\bar{Y}_t} \right)^{b_r} + Z_{n,t},$$

where ℓ_n captures innovation efficiency, $T_{n,t}$ generates knowledge spillovers from technologies already used in the country, $\bar{Y}_t$ is world output, and $b_r \in (0, 1)$ generates diminishing returns.

The innovator chooses $H_{n,t}^r$ to maximize

$$\Delta Z_{n,t} V_{n,t} - P_{n,t} H_{n,t}^r.$$

The first-order condition is

$$P_{n,t} H_{n,t}^r = b_r \Delta Z_{n,t} V_{n,t}.$$

- **Interpretation.** Innovation is more attractive when the value of a new technology $V_{n,t}$ is high.
- **Key intuition.** Better IP protection raises expected royalty income, which raises $V_{n,t}$ and therefore increases innovation.

3.5 Adoption and international technology licensing

A country- i adopter that tries to make a prototype from country n usable in production invests $h_{in,t}^a(j)$ and succeeds with probability

$$\varepsilon_{in,t}(j) = \bar{\varepsilon}_{in} \left(\frac{h_{in,t}^a(j)}{Y_t} \right)^{b_a},$$

with $b_a \in (0, 1)$.

The value of a successfully adopted technology for the innovator is

$$V_{in,t}^{\text{innov}}(j) = x_{in,t} \Pi_{in,t}(j) + \frac{P_{i,t}}{R_t P_{i,t+1}} V_{in,t+1}^{\text{innov}}(j),$$

while the value of an unadopted technology to the innovator is

$$J_{in,t}^{\text{innov}}(j) = \frac{P_{i,t}}{R_t P_{i,t+1}} \left[\varepsilon_{in,t}(j) V_{in,t+1}^{\text{innov}}(j) + (1 - \varepsilon_{in,t}(j)) J_{in,t+1}^{\text{innov}}(j) \right].$$

For the adopter, the value of a successful adoption is

$$V_{in,t}^{\text{adopt}}(j) = (1 - x_{in,t}) \Pi_{in,t}(j) + \frac{P_{i,t}}{R_t P_{i,t+1}} V_{in,t+1}^{\text{adopt}}(j),$$

and the value of an unsuccessful attempt is

$$J_{in,t}^{\text{adopt}}(j) = -P_{i,t} h_{in,t}^a(j) + \frac{P_{i,t}}{R_t P_{i,t+1}} \left[\varepsilon_{in,t}(j) V_{in,t+1}^{\text{adopt}}(j) + (1 - \varepsilon_{in,t}(j)) J_{in,t+1}^{\text{adopt}}(j) \right].$$

The law of motion for adopted technologies is

$$A_{in,t+1} = \varepsilon_{in,t}(Z_{n,t} - A_{in,t}) + A_{in,t},$$

and the total number of technologies used in country n is

$$T_{n,t} = \sum_{i=1}^M A_{ni,t}.$$

- **Interpretation.** Innovation creates blueprints; adoption turns them into usable production technologies.
- **Key channel.** Royalty payments are the observable counterpart of the model's licensing flows.
- **Why adoption matters dynamically.** A reform that raises royalty fees may discourage adoption in the short run even while stimulating innovation in the long run.

3.6 Royalty bargaining and the role of IP enforcement

Once a technology is successfully adopted, the innovator and adopter bargain over the royalty share $x_{in,t}$. The paper writes the Nash bargaining problem as

$$(x_{in} W_{in,t}(j) - 0)^{r_{in}} ((1 - x_{in}) W_{in,t}(j) - P_{i,t-1} h_{in,t-1}^a(j) - O_{in,t}(j))^{1-r_{in}},$$

where r_{in} is the innovator's bargaining power.

The key reduced-form link is

$$x_{in,t} = \bar{r}_{in} h_i y_{in,t},$$

where h_i is country i 's underlying IP-enforcement quality and $y_{in,t}$ is the policy variable chosen in the trade agreement.

- **Interpretation.** Weak IP enforcement means innovators do not receive the full royalty share they would get under perfect enforcement.
- **Why this matters.** A higher h_i or $y_{in,t}$ transfers more rents to innovators, raises the return to R&D, and increases long-run growth.
- **But there is a trade-off.** Higher royalties increase the cost of adopting foreign technologies, which can lower adoption and consumption in the short run for the reforming country.

3.7 Market clearing and feasibility

Output is allocated to consumption, innovation, and adoption:

$$Y_{n,t} = C_{n,t} + H_{n,t}^r + \sum_{i=1}^M H_{ni,t}^a.$$

Government tariff revenue rebated to households is

$$IBT_{n,t} = \sum_{i \neq n} \frac{t_{ni,t}}{1 + t_{ni,t}} p_{ni,t} P_{n,t} Y_{n,t}.$$

The world bond market clears:

$$\sum_{n=1}^M B_{n,t} = 0.$$

- **Interpretation.** The model closes like a standard dynamic general equilibrium trade model, but now with endogenous growth through licensing and innovation.

3.8 Trade-agreement bargaining problem

Countries i and n choose tariffs and IP enforcement to maximize a Nash product of welfare gains:

$$\max_{t_{ni,t}, y_{in,t}} \Delta \widetilde{W}_i(t, y)^v \Delta \widetilde{W}_n(t, y)^{1-v},$$

subject to

$$\Delta \widetilde{W}_i > 0 \quad \text{for both negotiating countries.}$$

Welfare gains are computed in consumption-equivalent units, inclusive of the transition:

$$\sum_{t=0}^{\infty} \beta^t u \left(C_{i,t}^*(t_{ni,0}, y_{in,0}) \left(1 + \frac{\Delta \widetilde{W}_i}{100} \right) \right) = \sum_{t=0}^{\infty} \beta^t u(C_{i,t}(t_{ni,t}, y_{in,t})).$$

- **Interpretation.** The agreement is cooperative, requires both countries to gain, and values welfare **along the transition**, not just at the final BGP.
- **Key modeling choice.** Under the baseline, the agreement is unexpected, permanent, and perfectly enforced.

3.9 Balanced growth path

The BGP growth rate satisfies

$$g^* \hat{T}^* = \Delta(g^*) \hat{T}^*,$$

so the common growth rate is pinned down by the interaction of innovation intensities and cross-country technology diffusion.

- **Interpretation.** All countries share a common long-run growth rate on the BGP, but they differ in relative levels.
- **Policy implication.** Tariffs and IP reforms affect both **levels** and **growth** by changing innovation and adoption incentives.

3.10 Gravity equations used for calibration

Trade flows imply a reduced-form gravity equation:

$$X_{in,t} = \exp(\beta^{RTA} RTA_{in,t} + f e_{n,t} + m_{i,t} + k_{in}) \varepsilon_{in,t},$$

where exporter-time, importer-time, and bilateral fixed effects recover trade costs and relative productivity.

Royalty flows imply

$$RP_{in,t} = \bar{r}_{in} h_i \frac{\varepsilon_{in}}{\varepsilon_{in} + g} \ell_n T_{n,t} \left(\frac{H_{n,t}^r}{\bar{Y}_t} \right)^{b_r} \frac{\Pi_{i,t}}{T_{i,t}},$$

which the paper rewrites as

$$RP_{in,t} = \exp \left(\sum_{k \in \{T, NT\}} \delta_k RTA_{in,t}^k + S_{n,t} + F_{i,t} + f e_{in} \right) u_{in,t}.$$

To infer IP enforcement, the importer-time fixed effect is regressed on GDP and the patent-rights index:

$$F_{i,t} = \beta_0 \log(GDP_{i,t}) + \beta_1 \log(GP_{i,t}) + m_{i,t}.$$

- **Interpretation.** Trade gravity pins down trade costs and productivity; royalty gravity pins down adoption rates and IP-enforcement quality.
- **Novel feature.** Royalty-payment data are used directly to discipline the technology-diffusion block.

3.11 Mechanism for transition dynamics

The short-run dynamics can be summarized by two first-order conditions emphasized in the paper:

$$H_{n,t}^r = \left(\frac{b_r \bar{Y}_t^{b_r} \ell_n T_{n,t} V_{n,t}}{P_{n,t}} \right)^{1/(1-b_r)},$$

and

$$H_{CHN,USA,t}^a = \left(\frac{b_a \bar{\varepsilon}_{CHN,USA}}{P_{CHN,t}} (V_{CHN,USA,t}^{\text{adopt}} - J_{CHN,USA,t}^{\text{adopt}}) \right)^{1/(1-b_a)}.$$

- **Interpretation.** Higher royalties raise the value of innovation but lower the value of adoption for Chinese users of foreign technology.
- **Core trade-off.** China can lose in the short run because adoption becomes more expensive before the higher innovation rate has time to raise growth enough to compensate.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- **Static analysis is incomplete.** A BGP-to-BGP comparison misses the fact that stronger IP protection requires costly transitional reallocations from adoption toward innovation.
- **Tariff-only or IP-only reasoning is misleading.** Lower tariffs without IP reform can benefit China in the short run but harm US innovators and long-run growth; IP reform without tariff relief can impose large transitional costs on China.
- **So the relevant object is dynamic welfare with transition.** This is exactly why the bargaining problem is defined on consumption-equivalent welfare inclusive of the transition.

4.2 Why royalty payments identify licensing and IP enforcement

- **Empirical motivation.** The paper first shows that royalty payments rise much more after RTAs with IP provisions than after RTAs without such provisions.
- **Structural discipline.** In the model, royalties are the observable manifestation of cross-country licensing. This makes royalty data informative about both **adoption probabilities** and the **effective royalty share innovators actually receive**.
- **Key identification move.** Bilateral fixed effects in the royalty gravity help recover diffusion terms, while importer-time effects combined with patent-rights data help recover the quality of IP enforcement.

4.3 Why trade gravity identifies trade costs and relative productivity

- **Trade shares depend on a small number of sufficient statistics.** Given the Armington block, bilateral trade flows map into trade costs and country productivity terms.

- **Exporter-time fixed effects.** These absorb country productivity and scale terms.
- **Pair fixed effects.** These absorb time-invariant bilateral frictions, letting the model recover trade costs in a gravity-consistent way.
- **Validation.** The estimated trade costs are negatively related to actual trade flows, and estimated productivity is strongly positively related to GDP per capita.

4.4 What calibrates the dynamic parameters

- **Growth target.** The paper targets an initial BGP growth rate of **1.85 percent**.
- **Innovation block.** The parameters b_r and ℓ_n are chosen to match the growth target and R&D intensity.
- **Adoption block.** The paper sets $b_a = b_r$ and uses royalty-flow estimates to pin down $\bar{\varepsilon}_{in}$.
- **Domestic adoption anchor.** It imposes $\varepsilon_{ii} = 0.5$, meaning domestic adoption takes about two years.
- **Royalty bargaining anchor.** The baseline bargaining component $\bar{r}_{in}$ is disciplined by the **25 percent rule** used in licensing practice.

4.5 Why the counterfactuals are informative

- **Same model, different policy bundles.** The paper changes tariffs and IP protection inside one calibrated equilibrium environment.
- **Perfect foresight transition.** This lets the model quantify how welfare accrues over time and not merely in the long run.
- **Alternative scenarios reveal mechanisms.** Comparing baseline, tariff-only, IP-only, unilateral reform, Nash equilibrium, and deviation-retaliation scenarios isolates what each policy instrument is doing.

4.6 Relation to the literature

- **Political economy of trade agreements.** The paper adds dynamic trade-and-growth structure to the literature on tariff bargaining and deep agreements.
- **Trade, innovation, and diffusion.** It extends modern quantitative trade-growth models by allowing negotiated tariffs and endogenous IP reforms.
- **IP policy.** Relative to the IPR-growth literature, the contribution here is to study bilateral agreements and their transition dynamics using licensing data.

5 Tables and figures

5.1 Figure 1: Dynamics of royalty payments around RTAs with IP provisions

Read:

- Royalty payments from developing to developed countries rise sharply after an RTA with **IP provisions** enters into force.
- The increase is much weaker for RTAs **without** IP provisions.
- This is the paper's empirical motivation for treating royalties as the main observable measure of licensing and technology transfer.
- It also supports the modeling assumption that improving IP protection inside trade agreements raises licensing flows.

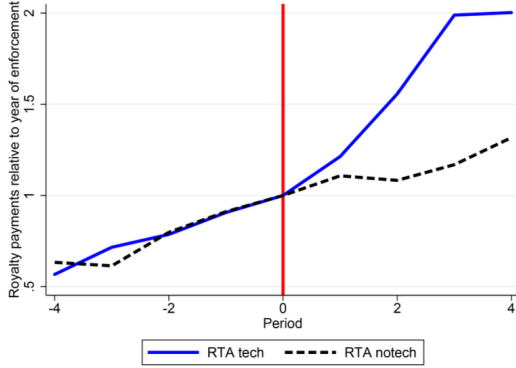


FIG. 1.—Dynamics of international technology licensing during RTAs with IP provisions. The figure shows the evolution of royalty payments from developing to developed countries 5 years before and 5 years after they sign a trade agreement with only technology provisions (solid line) and with only nontechnology provisions (dashed line). It considers all trade agreements signed between 1995 and 2012. The vertical line at zero represents the time at which the agreement enters into force.

TABLE 1
CALIBRATED PARAMETERS

Parameter	Value	Source
$\Omega_{US} (T_{US})^{1/\sigma-1}$	18.6	Gravity trade
$\Omega_{ROW} (T_{ROW})^{1/\sigma-1}$	5.98	Gravity trade
$\Omega_{China} (T_{China})^{1/\sigma-1}$	1.00	Gravity trade
$d_{US,ROW} (1 + \tau_{US,ROW})$	1.95	Gravity trade
$d_{US,China} (1 + \tau_{US,China})$	1.80	Gravity trade
$d_{ROW,USA} (1 + \tau_{ROW,USA})$	2.48	Gravity trade
$d_{ROW,China} (1 + \tau_{ROW,China})$	2.15	Gravity trade
$d_{China,USA} (1 + \tau_{China,USA})$	3.23	Gravity trade
$d_{China,ROW} (1 + \tau_{China,ROW})$	2.53	Gravity trade
L_{US}/L_{China}	.23	CEPII
L_{ROW}/L_{China}	1.82	CEPII
$\varepsilon_{US,ROW}$	.28	Gravity royalties
$\varepsilon_{US,China}$	.33	Gravity royalties
$\varepsilon_{ROW,USA}$	.34	Gravity royalties
$\varepsilon_{ROW,China}$	.15	Gravity royalties
$\varepsilon_{China,USA}$	.28	Gravity royalties
$\varepsilon_{China,ROW}$	.33	Gravity royalties
β_r	.52	Match $g = 1.85\%$
β_s	.52	Set $\beta_s = \beta_r$
λ_{US}	.40	Match R&D intensity in United States
λ_{ROW}	.50	Match R&D intensity in rest of world
λ_{China}	.18	Match R&D intensity in China
ρ_{in}	.25	Royalty fee
η_{US}	1.00	IP enforcement in United States
η_{China}	.40	IP enforcement in China
η_{ROW}	1.00	IP enforcement in rest of world

5.2 Table 1: Calibrated parameters

Read:

- The calibration is centered on the United States, China, and a residual rest of the world in the year 2000.
- China is calibrated as the least productive bloc in normalized terms, while the United States is the technology leader.
- A central number is $h_{China} = 0.4$, which implies that Chinese adopters pay only a fraction of the bargained royalty fee in the initial BGP.
- The estimated diffusion parameters imply that foreign technologies are adopted in roughly **three years on average**, while domestic adoption is anchored at about **two years**.

5.3 Figures 2 and 3: Validation of the gravity estimates

Read:

- Figure 2A shows that estimated trade costs are strongly negatively related to observed trade flows.

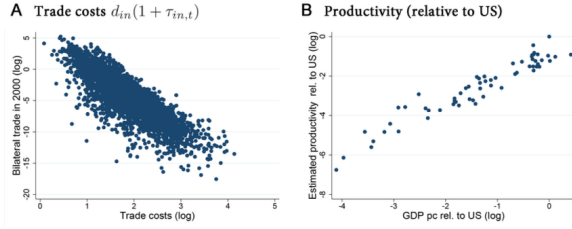


FIG. 2.—Estimated trade costs and productivity using gravity methods. A, Trade flows in data and trade costs from estimating gravity equation (30) with PPML methods and pair-fixed effects. B, Relative productivity estimated from exporter $\times$ time fixed effect and GDP per capita in data (relative to United States).

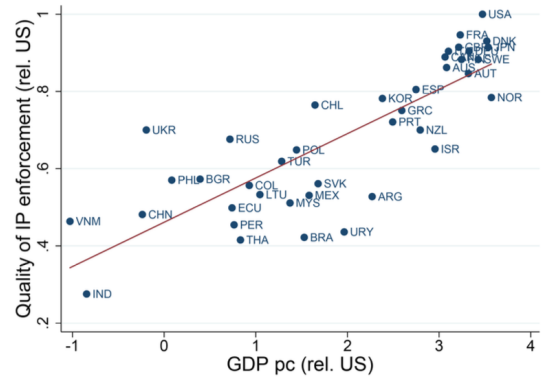


FIG. 3.—Quality of IP enforcement and GDP per capita relative to United States, averaged over period prior to 2000.

- Figure 2B shows that estimated relative productivity is strongly positively correlated with GDP per capita relative to the United States.
- Figure 3 shows that richer countries tend to have better estimated IP enforcement.
- These figures validate the structural use of trade and royalty gravity to discipline key country-level objects in the model.

5.4 Table 2: Baseline welfare gains from the trade agreement

Read:

- In the baseline cooperative agreement, the United States removes tariffs on Chinese imports and China raises the royalty fee paid to foreign innovators from **10 percent to 18 percent** while raising the domestic royalty fee to **25 percent**.
- Welfare rises in both countries, but more in the United States: about **0.853 percent** for the United States and **0.262 percent** for China in consumption-equivalent terms.
- The long-run growth rate rises from **1.85 percent to 1.87 percent**.
- This table establishes the paper's main quantitative result: deep agreements can raise global welfare and growth even if the developing country loses initially.

	$\Delta W(\text{USA})$ (1)	$\Delta W(\text{USA})$ (2)	g (3)	$\tau_{\text{USA,China}}$ (4)	$\xi_{\text{China,USA}}$ (5)	$\xi_{\text{China,China}}$ (6)
Baseline (%)	.85	.26	1.87	0	18	25

NOTE.—The table reports welfare gains, inclusive of the transition, computed from eq. (28) for the United States and China. Columns 1 and 2 display the welfare gains in the United States and China, respectively. Column 3 provides the BGP growth rate. Columns 4–6 contain the values of the policy instruments selected within the agreement: namely, tariffs, the royalty fee paid to foreign innovators, and the royalty fee paid to domestic innovators.

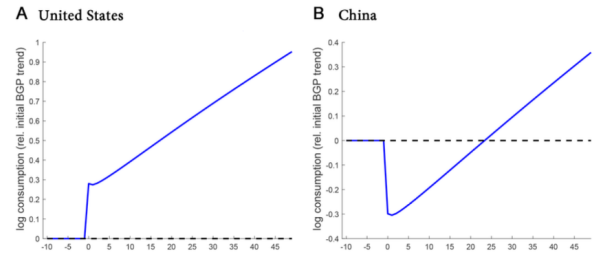


FIG. 4.—Log of consumption. The figure plots the log of consumption relative to its initial BGP trend in the United States (A) and China (B) 10 periods before and 50 periods after signing a trade agreement with IP provisions. The agreement is signed in period 1.

5.5 Figure 4: Consumption along the transition

Read:

- The United States experiences **short-run and long-run gains** after the agreement.
- China experiences a **drop in consumption on impact** and only later benefits from the higher growth path.
- The paper emphasizes that it takes roughly **25 years** for China to cross into positive net gains relative to the initial BGP path.
- This figure is the main visual proof that transition dynamics are first-order in the evaluation of IP provisions.

5.6 Figure 5: Productivity growth and relative productivity

Read:

- Productivity growth in both countries rises on impact, overshoots, and then converges to a higher common BGP growth rate.
- The agreement raises long-run growth through stronger incentives to innovate globally.
- Relative productivity of the United States versus China rises, so the agreement can increase **cross-country inequality in levels** even as it increases welfare everywhere.
- This is an important nuance: faster growth does not mean immediate convergence.

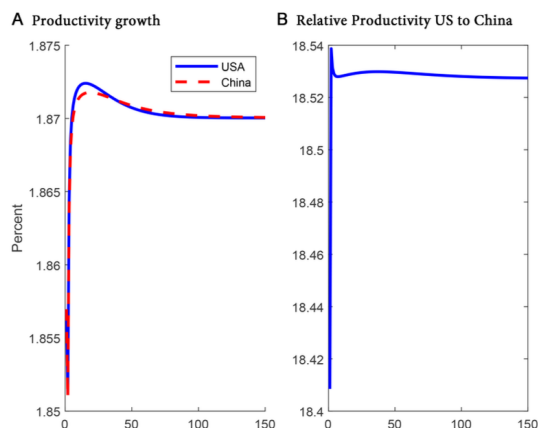


FIG. 5.—Growth rate of productivity. The figure plots the evolution of productivity growth in the United States and China (A) and relative productivity of the United States with respect to China (B) during the 150 years following the signature of a trade agreement with IP provisions designed as Nash bargaining. Period 0 represents the initial BGP.

TABLE 3
WELFARE GAINS: ALTERNATIVE SCENARIOS

Counterfactual	ΔW (USA) (%) (1)	ΔW (China) (%) (2)	BGP Growth (%) (3)	$\tau_{USA,China}$ (%) (4)	$\xi_{China,USA}$ (%) (5)	$\xi_{China,China}$ (%) (6)
A. Interactions between IP Reforms and Trade Policy						
1. Baseline	.853	.262	1.87	0	18	25
2. Only tariffs	-.097	.167	1.85	0	0	0
3. Only IPR	.944	.085	1.86	5	18	25
4. Unilateral IP reform	.117	.307	1.85	5	0	25
B. Revisiting Main Assumptions of Trade Agreement						
5. Nash equilibrium	.174	-.528	1.85	20	0	25
6. Shortsighted government	.270	.354	1.86	1	12	25
7. China deviates	.096	.316	1.85	5	0	25
8. China deviates (retaliation)	.228	-.080	1.85	20	0	25
9. Anticipated	.651	.258	1.86	5	16	25
10. Gradual	.657	.277	1.86	5	16	25

NOTE.—The table reports welfare gains inclusive of the transition, the BGP growth rate, and the terms of the agreement for alternative scenarios: (1) baseline, (2) only lower tariffs, (3) only IPR reform, (4) unilateral improvement of domestic IPR, (5) Nash equilibrium, (6) a shortsighted government, (7) China deviates without retaliation, (8) China deviates with retaliation, (9) anticipated policy, and (10) anticipated and gradual adjustment. In cases 5, 6, 9, and 10, I recompute the Nash bargaining solution from eq. (34). Columns 1 and 2 display the welfare gains in the United States and China, respectively. Column 3 provides the BGP growth rate. Columns 4–6 contain the values of the policy instruments selected within the agreement: namely, tariffs, the royalty fee paid to foreign innovators, and the royalty fee paid to domestic innovators. The values in the royalty fee columns represent the new royalty fees after the policy changes in each scenario.

5.7 Table 3: Alternative scenarios and what each instrument does

Read:

- **Only tariffs:** eliminating US tariffs without Chinese IP reform gives China gains but causes the United States to lose, and long-run growth remains lower than in the baseline.
- **Only IPR reform:** improving IP protection without tariff relief leaves China with only tiny overall gains because short-run adoption costs dominate.
- **Unilateral domestic IP reform:** China wants to strengthen domestic IP protection even without a bilateral agreement.
- **Nash equilibrium:** without cooperation, the United States raises tariffs to **20 percent**, China improves only domestic IP, China's welfare falls by about **0.528 percent**, and long-run growth stays at **1.85 percent**. Cooperation matters a lot.

5.8 Table 3 (continued): shortsighted government, anticipation, and gradualism

Read:

- A shortsighted government designs an agreement with smaller foreign-IP reform and only a modest tariff cut, which removes China's short-run pain but lowers long-run gains.
- In the anticipated and gradual scenarios, agents adjust before implementation, so the transition is smoother.
- The **gradual anticipated agreement** yields slightly higher welfare for China than the baseline because it reduces short-run losses while preserving higher growth.
- So not only the final policy package but also **the speed and timing of implementation** matter quantitatively.

TABLE 3
WELFARE GAINS: ALTERNATIVE SCENARIOS

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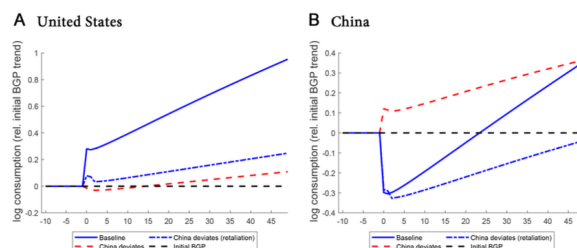


FIG. 6.—Log of consumption relative to initial BGP trend: China deviates. The figure plots the evolution of the log of consumption relative to its initial BGP trend in the United States (A) and China (B) 10 periods before and 50 periods after signing a trade agreement with IP provisions in the baseline (solid line), when China deviates two periods after the agreement entering in force and the United States goes back to the initial tariffs of 5% (dashed line), and when China deviates two periods after the agreement entering in force and the United States retaliates, imposing high tariffs of 20% (dotted line). The agreement is signed in period 1 and enters in force in period 11.

5.9 Figure 6: China deviates and the role of retaliatory tariffs

Read:

- If China deviates by cutting foreign royalty payments back down while the United States merely returns tariffs to their initial level, China gains and the United States loses relative to the baseline.
- If the United States retaliates by raising tariffs to **20 percent**, China suffers substantial losses.
- This figure provides the paper’s commitment logic: the agreement can be sustained if retaliatory tariffs are credible enough.
- In other words, tariffs are useful not only as bargaining chips ex ante but also as **discipline devices** ex post.

6 Short summary

- **Conclusion.** The paper develops a quantitative theory of bilateral trade agreements with IP provisions in which tariffs and IP reforms jointly shape trade, licensing, innovation, and welfare.
- **Main identification insight.** Trade gravity, royalty gravity, and patent-rights data together allow the model to infer trade costs, productivity, adoption rates, and effective IP enforcement.
- **Main economic lesson.** Strengthening IP protection can be growth enhancing, but the transition is politically difficult because the developing country may bear short-run losses unless tariff concessions or gradual implementation soften the blow.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Deep trade agreements that include IP provisions can generate **dynamic gains** by increasing royalty payments, stimulating innovation, and raising long-run growth.
- But those gains need not be evenly distributed over time. The country asked to strengthen IP enforcement can face immediate losses because adoption becomes more expensive before growth benefits arrive.
- A cooperative bargain over tariffs and IP enforcement dominates a noncooperative outcome in this environment.

7.2 Economic intuition

- Weak IP protection acts like a tax on innovation incentives: innovators are not fully paid for the technologies they create.
- Stronger IP protection fixes that distortion, but it also redistributes surplus away from adopters and toward innovators.
- Lower tariffs can compensate the reforming country by increasing access to markets and profits, thereby making foreign-IP reform politically feasible.
- Because innovation and adoption are forward-looking activities, all of this plays out through a long transition rather than instantaneously.

7.3 Takeaway

This paper is best understood as a dynamic political-economy model of deep trade agreements. Its central message is that tariffs and IP rules are complements inside trade negotiations: tariff concessions can buy stronger foreign IP protection, and stronger IP protection can raise world growth through licensing and innovation. But the same agreement that is efficient in the long run can be painful in the short run for the developing country. That is why transition dynamics, credible commitment, and gradual implementation are not side issues here—they are the core economics of the paper.